Orbital History Writes the Cosmic Clock on Dwarf Galaxies: A Pre-Registered, Modified-Gravity-Impossible Test of de Sitter–Unruh Modified Inertia

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Abstract

I pre-register a sharp, falsifiable prediction of the de Sitter–Unruh modified-inertia framework — the proposal that a body's inertia is its **nonlocal-in-time response to the cosmic-horizon Unruh bath**, a bath that carries exactly one clock, the de Sitter rate H_Λ, fixing a_0 = cH_Λ/Z = 9.36×10^{-11} m/s². Because inertia in this picture is a functional of the recent acceleration *history*, not of the momentary acceleration, the bath's clock must imprint on the internal dynamics of any object whose external acceleration *changes* on a timescale comparable to its own internal orbital frequency. The cleanest accessible site is a diffuse Local-Group dwarf galaxy on a radial-plunge orbit: at fixed pericenter distance and fixed mass, a high-eccentricity plunger should run **hotter** (larger internal velocity dispersion σ) than a circular-orbit dwarf of the same mass and same closest approach. The **sign is a theorem** (the memory kernel θ is decreasing, Milgrom-forced, so a plunge sheds adiabatic external loading and drops the dwarf deeper into the deep-MOND regime); the **magnitude is kernel-hostage** (+19–28% at the reference, a factor-~2 uncertainty riding on the undetermined θ(y)). The discriminating physical axis is **y = ω_ext/ω_internal**, gated by diffuseness, not raw eccentricity. The two named carriers that actually reach the non-adiabatic band are **Crater II (y = 3.28)** and **Antlia II (y = 2.55)**, against adiabatic controls Fornax and Sculptor (y < 0.3). Crucially, this prediction is **modified-gravity-impossible**: in any metric or field MOND (AQUAL/QUMOND/AeST) and in ΛCDM, the external-field effect is instantaneous — internal dynamics "depend only on the momentary value of a_ex" (Milgrom 2022) — so those theories predict *exactly zero* σ-vs-history correlation at fixed pericenter, for any a_0. Only modified inertia gives a nonzero, history-dependent σ; this joins the Cassini bound as a clean modified-inertia-vs-modified-gravity discriminator. **I report the honest current status straight.** A pre-registered pilot on 24 real Milky-Way dwarfs (Pace, Erkal & Li 2022, Gaia EDR3 orbits) gives a partial Spearman ρ(σ, eccentricity | r_peri, mass, r_half) = **−0.196, p = 0.395** — a **null**, slightly the wrong sign, robust to a tidal-heating control. This is **not** a falsification and **not** a hint: it is a null with almost no statistical power, because only two of 24 dwarfs reach the carrier band, eccentricity is a noisy surrogate for the real y-axis, and the per-object σ errors are comparable to the predicted signal. Existing data cannot yet test this prediction. The decisive test is Gaia DR4 (December 2026) plus diffuse-carrier spectroscopy and resolved profiles, run as a carrier-vs-control **y-contrast** with explicit tidal modeling. This is a pre-registered, near-term, MG-impossible prediction — **not a TOE, not a detection.**

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1. The framework as subject: inertia as a horizon-bath response

This paper treats one specific physical proposal as its subject and reasons forward from that proposal's own premises. The premise is not a variant of Milgrom's MOND and should not be read through MOND's lens.

The premise is this. An accelerated observer in de Sitter space sees a thermal bath — the Gibbons–Hawking/Unruh radiation of the cosmic horizon (Deser & Levin 1997). The framework takes the next step and asserts that **inertia itself is the body's reaction to that bath**: to resist acceleration is to push against the horizon's radiation. The bath has exactly one intrinsic rate, the de Sitter correlation frequency

$$H\_\Lambda = cH\_\Lambda/c = 1.81 \times 10^{-18}\ s^{-1},\ cH\_\Lambda = 5.42 \times 10^{-10}\ m/s^2,$$

set by the dark-energy density alone (the pure-Λ, de Sitter footing). The acceleration scale at which the bath response becomes order-unity is then

$$\mathbf{a_0 = cH\_\Lambda/Z = 9.36 \times 10^{-11}\ m/s^2,}$$

with Z the framework's geometric normalization. This is a *horizon-derived* acceleration, not a fitted parameter. Above a_0 a body is in the ordinary inertial regime; near and below a_0 the bath response weakens the effective inertia, and a fixed gravitational force produces a larger acceleration — the phenomenology usually labelled MOND, here arising from *modified inertia* with its own interpolation,

$$g\_obs = \sqrt{(g\_bar^2 + g\_bar \cdot a_0)},\ \mu\_fw(x) = (\sqrt{(1 + 4x^2)} - 1)/2x,$$

(the framework's own excess-heat interpolation; I use this throughout and never McGaugh's ν). On steady, hyperbolic-acceleration data (galaxy rotation curves, the radial-acceleration relation) this collapses algebraically to the same curve generic MOND produces, so the RAR cannot, by itself, distinguish the framework — a point I state plainly rather than spin. The framework's *distinctive* content lives where the acceleration history is non-trivial, which is exactly where this paper looks.

For the reader: the key word is **nonlocal-in-time**. Milgrom (1994) proved that any consistent modified-inertia theory must be time-nonlocal — the inertial reaction cannot be a function of the instantaneous acceleration alone; it must be a functional of the worldline carrying a memory kernel. The bath supplies the only clock that kernel can use: 1/H_Λ. A body's inertia therefore reads the *history-averaged* state of its motion. That is the engine of everything below.

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2. The prediction: a plunge runs hotter

Consider a diffuse dwarf galaxy orbiting the Milky Way. Two frequencies compete:

- ω_internal — the dwarf's own internal orbital frequency (its stars circulating in its potential), and
- ω_external — the rate at which the *external* (host) acceleration the dwarf feels is changing as it moves along its orbit.

Define the dimensionless ratio

$y = \omega\_\text{external} / \omega\_\text{internal}$

(Milgrom 2022, Eqs. 28/33/34). The memory kernel enters as $\theta(y)$:

- For a **circular** orbit, $|a\_\text{ext}|$ is constant. The kernel sees a stationary history. The response is purely adiabatic — the constant-external-field limit $\theta(0)$, which is the static external-field-effect "thermometer" reading already used for Crater II in earlier work.
- For a **radial plunge**, $|a\_\text{ext}|$ sweeps up as the dwarf falls in. Near pericenter ω_ext climbs toward ω_internal, so $y \longrightarrow O(1)$: the response is genuinely **non-adiabatic**, and $\theta(y)$ is a real, varying function rather than a constant.

Now the sign. The kernel θ is **decreasing** — $\theta(0)$ is a few, $\theta(1) = 1$ — a feature forced by Milgrom's construction, not chosen. A circular dwarf carries the full adiabatic external loading $\theta(0)\cdot g\_\text{ext}$, which partially pulls it *out* of the deep-MOND regime (this is the ordinary external-field effect that makes satellites a little more Newtonian). A plunging dwarf, momentarily non-adiabatic at pericenter, **sheds** part of that adiabatic loading: θ(y < the adiabatic value) means less effective external field, so the dwarf drops *deeper* into deep-MOND, its effective inertia falls further, and its stars move faster. **A plunge runs hotter.** Since σ scales as roughly the quarter-power of the effective regime depth,

σ_radial / σ_circular ≈ $(\theta(0)/\theta(1))$^¼ ≈ 2^¼ … e^¼ ≈ **1.19–1.28** (+19–28%),

for $\theta(0)$ in the range 2…e. The **existence and sign of the effect are kernel-independent theorems** (they follow only from θ decreasing); the **+19–28% number is θ(y)-hostage** to a factor of about two, because the framework has not derived the exact kernel — the same open $\theta(y)$ that bounds every relational prediction in this program.

Why diffuseness gates it. y is large only when ω_internal is *small* — i.e. for diffuse, low-internal-frequency dwarfs. A dense dwarf has high ω_internal, so even a steep plunge keeps $y \ll 1$ and the response stays adiabatic: dense dwarfs are *immune* and serve as a built-in internal control. Milgrom himself flagged the relevant population: "in some dwarf satellites of the Milky Way and Andromeda we estimate ω_ex ~ ω_in." Those diffuse satellites are precisely the carriers. The axis of the test is therefore **diffuseness-gated y**, not raw eccentricity — a distinction that turns out to be decisive for interpreting the pilot (Sec. 5).

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3. The modified-gravity-impossibility theorem

What makes this prediction a *discriminator* rather than just another MOND signature is that no modified-gravity theory can produce it, even in principle.

In every metric or field formulation of MOND — AQUAL, QUMOND, and the relativistic completion AeST (the framework's only covariant home) — the external-field effect is **instantaneous**. Milgrom (2022) states it verbatim: a subsystem's internal dynamics "depend only on the momentary value of a_ex." The internal dynamics are governed by the value of the external field *right now*, with no memory of how the system arrived there. Therefore, comparing two dwarfs of identical mass at identical pericenter distance — one on a circular orbit, one on a radial plunge — a modified-gravity theory predicts **exactly the same internal σ**, hence

$\rho(\sigma, \text{orbital history} \mid r\_peri, \text{mass}) = \mathbf{0}$, for any a_0, in any metric/field MOND.

ΛCDM/CDM gives the same null for the same reason: a circular and a radial subhalo at matched current radius have the same internal dynamics modulo tidal history (which is the confound, addressed in Sec. 6, not the signal). **Only modified inertia — inertia as the nonlocal bath-response functional — yields a nonzero, history-dependent σ.** The logical structure is identical to the Cassini bound, which excludes the framework's modified-gravity cousins because *they* would imprint a trajectory/history dependence on solar-system orbits that the data forbid; here the *same* nonlocality appears as a *positive* prediction in a fresh, accessible observable. A zero result is consistent with all of modified gravity and CDM; a nonzero positive σ-vs-history correlation at fixed pericenter would be impossible for any of them and required by the framework.

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4. The carriers, the controls, and the diffuseness gate

Pre-specifying carriers on physics (diffuseness), not on σ, the per-dwarf y computation places only two objects firmly in the non-adiabatic carrier band:

dwarf	role	a_ext/a_0 @ peri	$y = \omega\_ext/\omega\_in$	regime
Crater II	carrier	0.46	**3.28**	NON-ADIABATIC
Antlia II	carrier	0.35	**2.55**	NON-ADIABATIC
Boötes I	control	0.33	0.21	adiabatic
Fornax	control	0.14	0.16	adiabatic
Sculptor	control	—	0.12	adiabatic
Draco	control	—	0.07	adiabatic

Crater II is the prime carrier: extraordinarily diffuse (half-light radius ~1.1 kpc, the lowest internal density in the sample), eccentric (e = 0.71), with a *large* pericenter (24 kpc). **Antlia II** is the largest-radius dwarf known (~2.9 kpc), low density, eccentric (e = 0.56), pericenter 38 kpc. Both reach y ~ 2.5–3.3 at pericenter — squarely non-adiabatic. Both have large pericenters, which is a genuine design strength: they sit *away* from the tidal-stripping regime, so the prediction's two real carriers are **tide-clean by construction** (Sec. 6). Everything denser — Fornax, Sculptor, Draco, the compact ultra-faints — stays adiabatic ($y < 0.3$) and acts as the control locus. The sample is thus sharply **bimodal**: a 2-object carrier set against a ~22-object adiabatic background. This bimodality is the central fact governing what existing data can and cannot say.

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5. The pre-registered pilot: a null at low power, reported straight

Before any fitting, I pre-registered the test (the registration, sign convention, primary statistic, and the prime confound are recorded in the data script; see Reproducibility). The hypotheses:

- **H1 (framework):** at fixed pericenter and mass, residual internal σ correlates **positively** with orbital eccentricity, predicted amplitude +19−28% radial-vs-circular at matched pericenter.
- **H0 (modified gravity / CDM): zero** partial correlation.

Data are a single homogeneous source — Pace, Erkal & Li 2022 (ApJ 940, 136; Gaia EDR3 proper motions) for *all* σ_{los} and *all* orbital parameters — minimizing cross-catalog systematics. N = 24 usable Milky-Way dwarfs (both σ and eccentricity measured). The pre-specified primary statistic is the partial Spearman correlation of σ_{los} with eccentricity, controlling for pericenter, a luminosity-based mass proxy, and half-light radius.

Result. The point estimate is negative and insignificant in every variant:

variant	ρ	p (two-sided)	reading
PRIMARY partial $\rho(\sigma, \text{ecc} \mid r_{peri}, L, r_{half})$	**−0.196**	**0.395** (dof = 19)	null, wrong-signed
virial-deviation $\rho(\log \sigma, \text{ecc} \mid \sqrt{L/r_{half}}, r_{peri})$	−0.220	0.313	null
simple Spearman(σ, ecc)	−0.113	0.598	null
no-LMC eccentricities	−0.030	0.899	even flatter

The correlation is small, slightly **negative** (the *opposite* sign to the predicted positive effect), and statistically indistinguishable from zero (all p between 0.40 and 0.90). Adding the pre-registered tidal proxy $M_{MW}(<r_{peri})/r_{peri}^3$ as an extra control leaves it essentially unchanged: $\rho = -0.196$, $p = 0.408$. The result is **robust to the tidal-heating control** — but only because there is no signal for tides to displace.

I report this as what it is: a null with almost no power, not a strike against the theory. Three reasons, all physical rather than a matter of bad luck:

1. **The axis is wrong for existing data.** The framework's signal lives on **y**, gated by diffuseness; only **two** of 24 dwarfs (Crater II, Antlia II) reach the carrier band. A population partial correlation on *raw eccentricity* is a weak surrogate — with 2 carriers buried in 22 adiabatic controls it cannot deliver the carrier-vs-control y-contrast the prediction actually makes.
2. **The σ errors are comparable to the predicted signal** (~10–40%, and *largest* on the carriers — Antlia II is 5.71 ± 1.08 km/s). The +19–28% effect is at or below this floor per object.
3. **An artifact pushes the carriers slightly cold.** The carriers sit modestly below a naive luminosity/r_half virial baseline (Crater II σ_dev = −0.36, Antlia II −0.06). This is partly mechanical, not evidence against: y is itself built from σ (a low-σ diffuse dwarf is automatically high-y), so y and a σ-residual anti-correlate by construction. The clean, pre-registered eccentricity-axis test is the one to read — and it is null.

So: not suggestive (no surviving positive correlation), not tidal-confounded (robust to the control), simply **underpowered**. **Existing literature data does not yet show the effect, and existing data was never going to.** This pilot establishes only that fact; it does not, and cannot, move the theoretical status of the prediction. The sign remains a theorem.

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6. The decisive test and its one real confound

The right test is **not** a raw-eccentricity population correlation. It is a **carrier-vs-control y-contrast** with the discriminating axis measured per object. Three ingredients make it decisive, all arriving on a near-term timeline:

1. **Gaia DR4 (December 2026)** sharpens per-dwarf proper motions, hence pericenter and the orbital frequency ω_ext, tightening y object-by-object — the axis is *measured*, not statistically inferred (the decisive improvement over the earlier, low-purity cluster-UDG version of the same physics, where infall phase had to be guessed at ~0.4 purity).
2. **A larger diffuse-carrier set** — Antlia II, Crater II, plus newly-characterized large-radius low-density satellites from DES/LSST follow-up — beats down per-object noise by turning a 2-point contrast into a population.
3. **Resolved σ profiles** (Keck/VLT/MUSE multi-object spectroscopy) rather than a single global σ.

The one real confound is tidal heating. A radial plunger with a small pericenter is also tidally stirred, which would *itself* inflate σ and could mimic the framework's positive signal. The design separates the two:

- The prediction is the σ excess *at fixed pericenter* — tidal heating is controlled by regressing on pericenter (and r_peri/r_tidal where available).
- Tides predict a **different radial σ profile** and a host-aligned stripping/elongation morphology, separable with resolved kinematics — whereas the bath-response effect is a global re-deepening into deep-MOND with no preferred tidal axis.
- The two named carriers are **tide-clean by construction**: Crater II and Antlia II have large pericenters (24, 38 kpc), sitting away from the stripping regime, yet are non-circular. They are

non-adiabatic *without* being strongly tidally heated — exactly the clean corner the population test should target.

Run this way — carrier-vs-control y-contrast, Gaia DR4 orbits, resolved profiles, explicit tidal modeling — the test discriminates modified inertia from modified gravity at a sign level: a positive carrier-vs-control σ excess at matched pericenter is required by the framework and forbidden to all of metric/field MOND and CDM.

7. Scope and honest limitations

- **Pre-registered and falsifiable.** The hypotheses, sign convention, primary statistic, and prime confound were fixed before fitting. A null carrier-vs-control y-contrast at DR4 precision, or a *negative* one surviving the tidal control, would count against the framework's modified-inertia content.
- **Modified-gravity-impossible.** The signal's existence — not merely its size — is forbidden to every metric/field MOND and to CDM (Milgrom 2022). This is what makes it a discriminator, joining **Cassini** as a clean modified-inertia-vs-modified-gravity test.
- **The sign is a theorem; the magnitude is kernel-hostage.** Plunge ⟶ hotter follows from θ decreasing alone. The +19–28% number rides on the undetermined memory kernel θ(y), good to a factor of ~2 — the same open θ(y) that bounds every relational prediction in this program.
- **Near-term.** The decisive data (Gaia DR4) arrive December 2026; the carriers are named and the spectroscopy largely exists.
- **What this is not.** It is **not a TOE** — the framework does not derive the Standard Model and is a one-parameter effective theory at a frontier. It is **not a detection** — the present pilot is an explicit null at low power. It is **not a claim that existing dwarf data shows the effect** — they do not, and could not.

The honest standing: a sharp, MG-forbidden, pre-registered prediction with a measured discriminating axis and a named carrier set; a current null that reflects the data's lack of power rather than the theory's failure; and a decisive near-term test that the framework will pass or fail cleanly.

Reproducibility

All numbers in Sec. 5 reproduce from two scripts in `real_research/reviews/`:

- `dwarf_ecc_sigma_pilot_data.py` — the 24-dwarf data dictionary (every value carries a per-object source citation; σ_los and orbits from Pace, Erkal & Li 2022 / Gaia EDR3; Simon 2019 and McConnachie 2012 as cross-checks), the pre-registered hypotheses, sign convention, primary statistic, and the tidal confound control. Footing: $a_0 = 9.36 \times 10^{-11}$ m/s², framework interpolation only.
- `dwarf_ecc_sigma_pilot_analysis.py` — the per-dwarf y = ω_ext/ω_in computation and the partial-correlation test.

Supporting derivations: `real_research/NATIVE_CONSEQUENCES_HORIZON_INERTIA_2026-06.md` (the prediction's derivation from the bath premise) and `real_research/DWARF_ECC_SIGMA_PILOT_2026-06.md` (the full pilot verdict).

References (key): Milgrom 1994 (modified-inertia must be time-nonlocal); Milgrom 2022 (the kernel θ(y), and the instantaneous external-field effect of modified gravity); Deser & Levin 1997 (de Sitter–Unruh temperature); Pace, Erkal & Li 2022, ApJ 940, 136 (σ and Gaia EDR3 orbits); Simon 2019, ARA&A 57, 375; McConnachie 2012, AJ 144, 4.

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